

Teamwork Agreement

Image Based Manufacturing Inspection System

MathWorks Workplace Challenge Project

Project: Image-Based Defect Detection for Manufacturing Inspection

Team Name: Team 12

Instructor: Kevin Graham

Program/Class: Engineering Pathways

Members and Roles

1. **Member:** Christian, **Team Role:** Computer Vision Engineer (Modeling, Analysis, and Visualization)
2. **Member:** Destiny, **Team Role:** Computer Vision Engineer
3. **Member:** Carlos, **Team Role:** Computer Vision Engineer

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts.

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations

- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Group Chat, iMessages
- Secondary channel: Discord channel: Team 12
- Response time: within 8 hours on weekdays and 12 hours on weekends
- Meeting cadence: Weekly Thursday meetings through FaceTime
- Decision records: Major Decisions written on a Google Document and MATLAB Live Scripts

4) Collaboration & Tools

- Version control: Git and GitHub Repository
- Data management: All datasets, parameters, and assumptions documented in README and MATLAB Live Script (ImageBasedDefectSystem.mlx)

File naming & style: /src files - camel case, /scripts - snake case, MATLAB Code: camel case

- Backups: GitHub (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

```
28 % Select image to inspect
29 inputImage = readimage(imds,defectIndex);
30
31 [rgbImage, grayImage] = preprocessImage(inputImage);
32
33
34 % Part ROI Detection
35
36 % 1. Convert image to binary
37 BW = imbinarize(grayImage);
38
39 figure
40 imshow(BW)
41 title("Binary Image")
42
43 % 2. Invert so the screw is white
44 BW = ~BW;
45
46 figure
47 imshow(BW)
48 title("Inverted Binary Image")
49
50 % 3. Remove small objects
51 BW = bwareaopen(BW,50);
52
53 figure
54 imshow(BW)
55 title("Small Objects Removed")
56
```

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Assigned team tasks, established weekly meeting dates and times. Established main communication channels, made a project folder, and teammates cloned the repo. Started by studying all of the Onramp MATLAB recommended tutorials.
- Week 2: Finished uploading task 1, “Explore and Organize Image Dataset,” and task 2, “Build a Single-Image Inspection Function”.
- Week 3: We debugged tasks 1 and 2, and finished task 3, “Train and Integrate an AI Classifier into your Single-Image Inspection Function”. We also were able to complete the checkpoint, “Combined outputs into a hybrid inspection result”. Finished task 4: Evaluate Inspection System Performance
- Week 4: Finishing up organization of GitHub repository and uploading task 5, “Test and Evaluate System Robustness and Interpret Results”. We are also debugging

steps 3 and 4, documenting our Readme, and completing our MATLAB Live Script documentation.

7) Decision-Making & Conflict Resolution

- Default: Consensus with PM Tiebreaker (e.g. consensus after discussion; PM facilitates)
- If blocked >48 hrs: Group decision is made by voting, resulting in majority rule (e.g. Time-limited debate, majority vote, PM documents decision)
- Technical disagreements: Each member presents their arguments to the team, and a vote decides the direction of the team. (e.g. Evidence-based comparison before vote)
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: Google Docs task board for the week (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: If missed deadlines, members can be updated through the group chat. (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)
- Attendance: Give a heads up at most an hour ahead of time if a member is going to be late or can not make the meeting (e.g. If you'll miss a meeting, give 24-hour notice and share updates asynchronously.)

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Explore and Organize the Image Dataset
- Build a Single-Image Inspection Function
- Train and integrate an AI Classifier into the single-image inspection function.
- Evaluate Inspection and System Performance
- Test and Evaluate System Robustness
- Complete Template
- Complete README FILE
- Complete live MATLAB Script
- Debug and check spelling for all code and Documentation

(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: Carlos Tapia Date: 6-14-2026

Name & Signature: Christian Loriega Date: 6-14-2026

Name & Signature: Destiny Aguillon Date: 6-18-2026